MILITARY SPECIFICATION

ACTUATOR

* This amendment forms a part of Military Specification MIL-A-23282B(Wep) of 19 April 1965 and is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

By this amendment basic specification MIL-A-23282B (Wep) is changed to read MIL-A-23282B(OS).

* Wherever it appears in specification, delete "Bureau of Naval Weapons" and substitute "Naval Sea Systems Command"; also delete "BUWEPS" and substitute "NAVSEA".

Page 1, paragraph 1.1: Delete "Torpedo Mk 37 Mods 0 and 1" and substitute "Torpedo Mk 37 Mods 0, 0 (M) , 1, 2, and 3".

* Bottom of page 1, add the following:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station (611), Standardization Division, Indian Head, MD 20640 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Page 5, paragraph 3.2.7 (Interchangeability) : Delete.

* Page 6, paragraph 3.3.2.1. Delete and substitute:
"3.3.2.1 Clutch circuit resistance. Each magnetic particle clutch circuit shall have a dc resistance of 5720 ohms plus or minus 7½ percent, tested at 22 degrees centigrade plus or minus 3 degrees centigrade when tested as in 4.5.4.2."

Page 8, paragraph 3.3.4.1, line 8: Delete "6.7 volts or more than 8.1 volts" and substitute "5. 3 volts or more than 6.9 volts with a 1020 ± 20 ohm load".

Page 9, Figure 1: Replace figure with new Figure 1, attached.

Page 10, paragraph 3.3.6.1: Delete last four lines and substitute "... current of 7 plus or minus 2 milliamperes (ma) dc energizes the magnetic particle clutch coil across terminals 1 and 2 of P40 (Drawing 694739). Test as in 4.5.4.6.1".

FSC 1355

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:17Z
Check the source to verify that this is the current version before use.

Page 10, paragraph 3.3.6.2: Delete last four lines and substitute "... and a current of 7 plus or minus 2 ma dc energizes the magnetic particle clutch coil across terminals 2 and 3 of P40 (Drawing 694739). Test as in 4.5.4.6.2."

Page 13, Figure 2: Replace figure with new Figure 2, attached.

* Page 15, paragraph 4.1. Delete and substitute:
"4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

Page 16, paragraph 4.2.1, lines 4 and 5: Delete "Torpedo Mk 37 Mods 0 and 1 and substitute "Torpedo Mk 37 Mods 0, O(M), 1, 2, and 3".

Page 21. Add the following new paragraph:
"4.5.4.2.5 Following the assembly of the complete actuator, use the multimeter (4.5.3.4) to ensure that the wiring conforms to schematic diagram (Drawing 694739) and that the pin to pin resistances are as set forth in Table I.

Table I

RESISTANCE IN OHMS

Pins on P40	Output Shaft Fully Counterclockwise	*Output Shaft Moving Clockwise	Output Shaft Fully Clockwise
5-6	7,900 ±10%	Potentiometer used is to have a resolution of 0.15% max	2,100 ±10%
5-7	10,000 ±10%	10,000 ±10%	10,000 ±10%
6-7	2,100 ±10%	Potentiometer used is to have a resolution of 0.15% max	7,900 ±10%
9-10	490 ±10%	Potentiometer used is to have a resolution of 0.15% max	135 ±10%
9-11	625 ±10%	625 ±10%	625 ±10%
10-11	135 ±10%	Potentiometer used is to have a resolution of 0.15% max	490 ±10%
13-14	- -	Polarity 14 is + volts	- -

*During output, the shaft movement shall be continuous without interruption."

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:17Z
Check the source to verify that this is the current version before use.

Page 24, paragraphs 4.5.4.6.1 and 4.5.4.6.2. Delete and substitute:

"4.5.4.6.1 With the actuator mounted as in 4.5.4.4.1, the actuator adjustment set connected as in 4.5.4.4.2, and with the pulley driven at 5000 to 6000 rpm in a cw direction (as viewed facing the pulley), energize the magnetic particle clutch coil connected across terminals 1 and 2 of P40 with 7 plus or minus 2 ma. The output arm shall move as specified in 3.3.6.1.

4.5.4.6.2 With the actuator mounted and driven as in 4.5.4.6.1, energize the magnetic particle clutch coil connected across terminals 2 and 3 of P40 with 7 plus or minus 2 ma. The output arm shall move as specified in 3.3.6.2."

Page 31, paragraph 6.1: Delete "Torpedo Mk 37 Mods 0 and 1" and substitute "Torpedo Mk 37 Mods 0, 0 (M), 1, 2, and 3".

* After Page 31: Delete the old DD form 1426 and add the attached DD Form 1426 to end of specification.

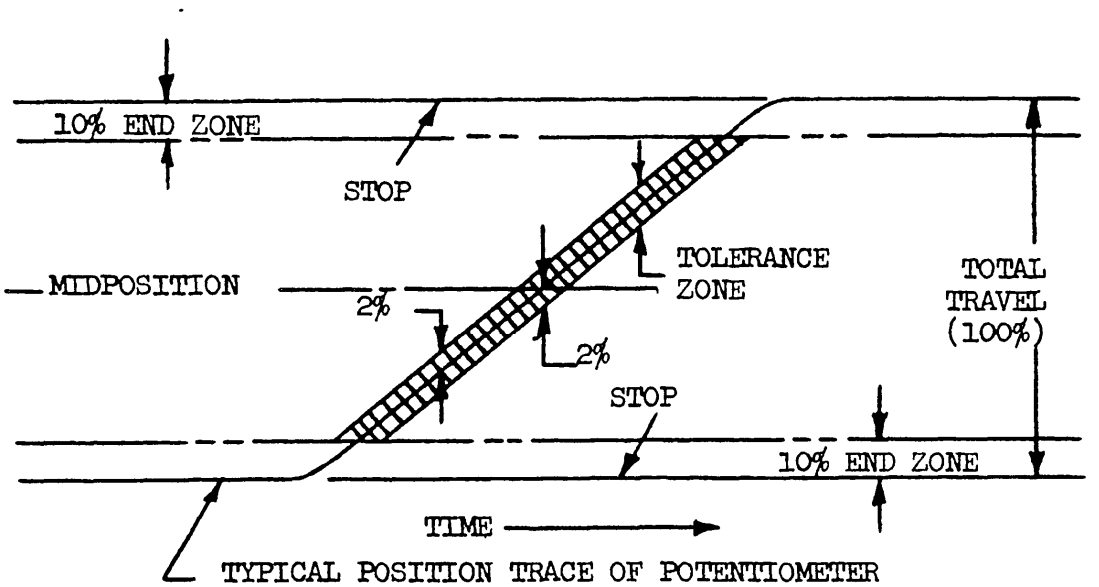
The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Navy - OS

Preparing Activity:
Navy - OS
Project No. 1355-N387

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:17Z
Check the source to verify that this is the current version before use.

LINEARITY & SMOOTHNESS OF OPERATION



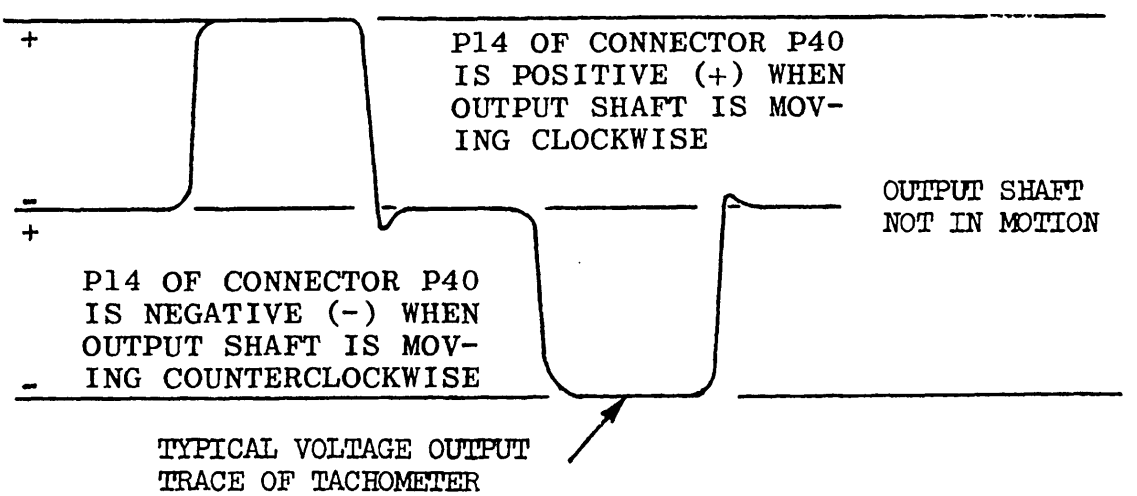
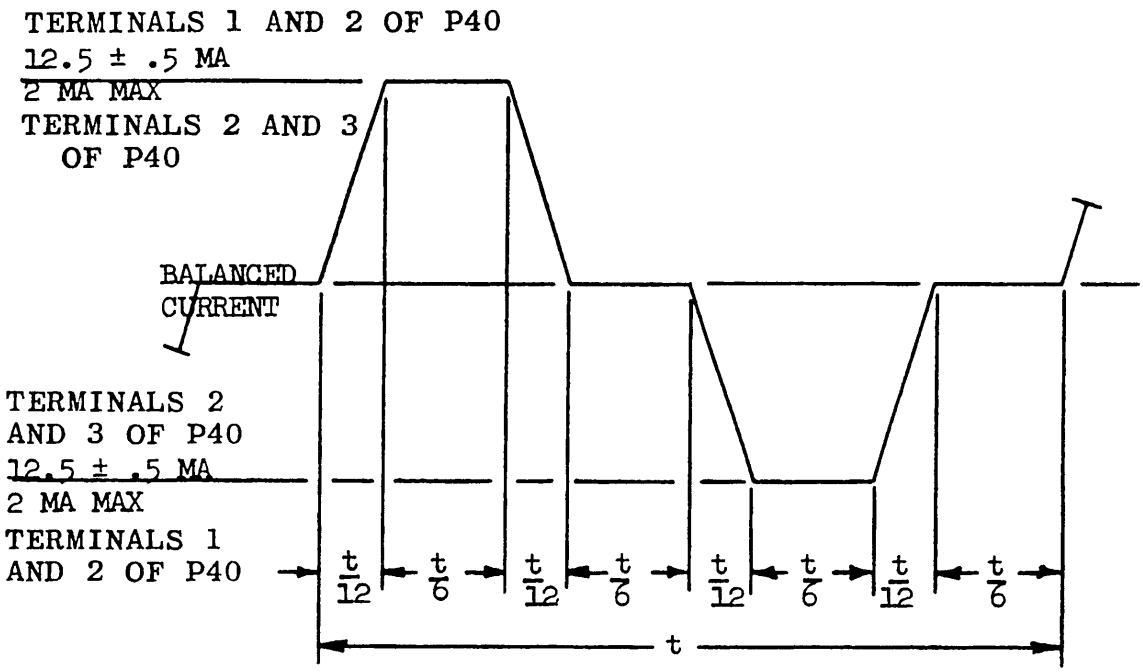


FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 20260-19T16:17Z
Check the source to verify that this is the current version before use.

DUTY CYCLE



ONE ELEMENT OF A DUTY CYCLE WHERE: t MAY VARY FROM 6 TO 150 SECONDS AND WHERE THE DUTY CYCLE γ IS 1800 SECONDS MINIMUM.

ANY INCREMENTAL TIME MAY VARY BY 25% OF ITS OWN VALUE.

THE NUMBER OF ELEMENTS, N, TO COMPLETE A DUTY CYCLE: $N = \frac{\gamma}{t} = \frac{1800(MIN)}{t}$

FIGURE 2

☆ U. S. GOVERNMENT PRINTING OFFICE: 1977-703-020/1458

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:17Z
Check the source to verify that this is the current version before use.